

Computer Science Tutorial — ENS Year 2 (EN)

****Instructor:**** BELACEL Madani

****Level:**** ENS — Year 2

****Duration:**** 1h30 (90 min)

Tutorial objective

Practice OOP, algorithms, SQL, Web, Git, and lesson plan design.

Exercise 1 — Python OOP *(18 min)*

****Task:**** Create a `Grade` class (subject, value, weight) and a `ReportCard` class computing weighted average.

****Detailed answer key:****

```
class Grade:
    def __init__(self, subject, value, weight=1):
        self.subject = subject
        self.value = value
        self.weight = weight
class ReportCard:
    def __init__(self, grades):
        self.grades = grades
    def average(self):
        s = sum(g.value * g.weight for g in self.grades)
        w = sum(g.weight for g in self.grades)
        return s / w
```

Exercise 2 — Binary search *(18 min)*

****Task:**** Binary search pseudocode. Precondition? Complexity?

****Detailed answer key:****

- ****Sorted**** array required.
- Divide and conquer: compare middle, shrink interval.
- Complexity $O(\log n)$.

Exercise 3 — SQL *(18 min)*

****Task:**** Tables `student(id, name)` and `grade(student\_id, subject, value)`. Query: average per student.

****Detailed answer key:****

```
SELECT s.name, AVG(g.value) AS average
```

```
FROM student s
```

```
JOIN grade g ON s.id = g.student_id
```

```
GROUP BY s.id, s.name;
```

Exercise 4 — Web page ***(18 min)***

****Task:**** HTML form (name, email) + centered CSS + JS alert on submit.

****Detailed answer key:****

- `<form>` with `input type="text"` and `email`.
- CSS `margin: auto; max-width: 400px`.
- `onsubmit` → `preventDefault()` + `alert('Thank you')`.

Exercise 5 — Git and lesson plan ***(13 min)***

****Task:**** Describe Git workflow for pair project. Sketch 45-min "Python loops" lesson for grade 9.

****Detailed answer key:****

- Branches `main` + `feature`; atomic commits; merge request.
- Lesson: 10 min review, 25 min guided exercises, 10 min challenge; criteria: working, readable code.

Session schedule — 90 min

Ex.1 18' | Ex.2 18' | Ex.3 18' | Ex.4 18' | Ex.5 13' | Closing 5'